

Rishika Pamanji

Research Scholar, University of Houston, Sugar Land, TX 77479, | rishikareddy135@gmail.com | (650)402-5392|
<https://www.linkedin.com/in/rishikapamanji/>

SUMMARY

PhD Researcher with experience in SQL, Python, and machine learning, specializing in large-scale biomedical data analysis, predictive modeling, and AI-driven research. Currently conducting fellowship-level research on biomarker discovery for Triple Negative Breast Cancer (TNBC) using advanced ML algorithms including SVM and XGBoost. Proven ability to build end-to-end data pipelines, interactive dashboards, and deploy production-grade ML models across healthcare and enterprise domains.

TECHNICAL SKILLS

Languages:	Python, SQL
Libraries:	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Sentence Transformers, LangChain
Machine Learning:	SVM, XGBoost, Random Forest, CNN, LSTM, NLP, Classification, Regression
Visualization:	Power BI, Tableau, Excel
Tools:	AWS, Snowflake, Microsoft Fabric, Jupyter Notebook, Git, Github, Streamlit, Obsidian
Databases:	MySQL, PostgreSQL, SQL Server
Vector Databases:	FAISS, Chroma
Generative AI & LLMs:	Large Language Models (LLMs), RAG, Ollama, Prompt Engineering

PROFESSIONAL EXPERIENCE

REALTECHINC, LLC May 2026 – Aug 2026 *AI/ML Intern, USA*

- Developed and tested domain-focused Large Language Model (LLM) workflows using Ollama, including prompt design, category normalization, deterministic output validation, and comparison with traditional machine learning approaches.
- Built Retrieval-Augmented Generation (RAG) and chatbot pipelines using Python, document chunking, embeddings, vector databases, semantic retrieval, and source-aware response generation for technical knowledge bases.
- Developed semantic search and text classification workflows using SentenceTransformer embeddings, cosine similarity, and Logistic Regression, processing 11,417 records through scalable batch pipelines.
- Benchmarked Logistic Regression configurations using saga and lbfgs solvers and 70:30, 80:20, and 90:10 train-test splits to evaluate model generalization, class representation, execution time, and reliability.
- Evaluated fine-tuned/domain LLM production readiness using accuracy, precision, recall, F1 score, confusion matrices, hallucination checks, output-format compliance, consistency, failure handling, latency, and category-level performance.
- Automated Excel-to-LLM workflows for batch question answering and structured output generation, while measuring processing time and validating model responses for large-scale execution.

ITion Solutions Inc. Aug 2023 – Dec 2025 *Data Analyst, USA*

- Designed and deployed Power BI dashboards with drill-through analytics, improving executive decision-making efficiency by **20% across business units**.
- Analyzed large-scale ERP datasets using SQL and Python (Pandas, NumPy), delivering scalable automated reporting pipelines that reduced manual reporting time by **40%**.
- Developed and fine-tuned ML models for text classification using Scikit-learn, reducing manual review workload by **30%** and increasing categorization accuracy.
- Automated end-to-end data pipelines integrating Snowflake and Microsoft Fabric, improving data ingestion workflow efficiency and ensuring high data quality.
- Collaborated with cross-functional stakeholders to translate business requirements into data

models and KPI frameworks using Agile methodology.

ITion Solutions Limited Apr 2019 – Jul 2021 *Data Analyst*, India

- Designed and maintained SQL pipelines supporting business intelligence operations, enabling near-real-time reporting across enterprise datasets.
- Performed Exploratory Data Analysis (EDA) on structured datasets and built Tableau dashboards used by senior management for monthly performance reviews.
- Improved overall data quality by implementing robust data cleaning, normalization, and validation frameworks, reducing downstream data errors by **25%**.
- Partnered with database administrators to optimize query performance, cutting average report generation time by **35%**.

Siemens Healthineers, May 2019 – Aug 2019, Internship, India

- Portal development for license management to molecular imaging's collaboration sites

RESEARCH & PROJECTS

AI-Based Biomarker Identification for Triple Negative Breast Cancer (TNBC) *Ongoing – PhD Research*

- Conducting fellowship-level research to identify novel genomic and proteomic biomarkers for TNBC, one of the most aggressive and clinically challenging breast cancer subtypes, using high-dimensional biomedical datasets.
- Implemented and benchmarked multiple supervised ML classifiers including **Support Vector Machines (SVM)** with RBF and linear kernels and **XGBoost** gradient boosting, achieving competitive AUC-ROC scores on imbalanced clinical datasets.
- Engineered a comprehensive **feature selection pipeline** combining filter methods (Chi-Square, ANOVA F-test), wrapper methods (Recursive Feature Elimination), and embedded methods (LASSO regularization) to isolate the most biologically relevant biomarker candidates from thousands of genomic features.
- Developed Python-based preprocessing modules for normalization, missing-value imputation, and SMOTE-based oversampling to handle class imbalance inherent in clinical cancer datasets.
- Applied dimensionality reduction techniques (PCA, t-SNE) to visualize high-dimensional feature spaces and validate cluster separability between TNBC and non-TNBC subtypes.
- Focused on translational early-detection strategies, with findings aimed at informing clinical screening protocols and potential therapeutic target identification.

Breast Tumor Image Classification Apr 2022

- Built deep learning models using **CNN and ResNet-50** architectures for binary classification of malignant vs. benign breast tumor histopathology images, achieving **92% accuracy**.
- Applied image augmentation techniques (rotation, flipping, zoom) and transfer learning to improve model generalization on limited medical imaging datasets.

Malware Detection Using Deep Learning Dec 2023

- Designed a **CNN-based malware detection model** trained on byte-level binary representations of executable files, achieving **98% accuracy** on a multi-class classification benchmark.
- Optimized model robustness using dropout regularization and data augmentation, significantly reducing false-positive rates in adversarial testing scenarios.

RESEARCH PUBLICATIONS

- Alipour, Shawheen, Rishika Pamanji, Emina Jamil, Suneetha Yeguvapalli, and Kumaraswamy Naidu Chitrala. "A Blood-Based Transcriptomic Signature for PTSD Classification Using Machine Learning." medRxiv (2026): 2026.05.10.26352854. Published May 13, 2026.

- Tusha Sharma, Rishika Pamanji, Suneetha Yeguvapalli, Raphael Amediku, Harsha Sri Kaki, Fatima Merchant, Kumaraswamy Naidu Chitralla. Impact of Olaparib and Atezolizumab on the transcriptome profile of Triple-Negative Breast Cancer Cells. *International Journal of Molecular Sciences*. Under communication [Impact Factor:5.6]
- Bommu, Uma Devi, Kranthi Kumar Konidala, **Rishika Pamanji**, and Suneetha Yeguvapalli. "Computational screening, ensemble docking and pharmacophore analysis of potential gefitinib analogues against epidermal growth factor receptor." *Journal of Receptors and Signal Transduction* 38, no. 1 (2018): 48-60. [Impact Factor: 2.3]
- Bommu, Uma Devi, Kranthi Kumar Konidala, **Rishika Pamanji**, and Suneetha Yeguvapalli. "Structural probing, screening and structure-based drug repositioning insights into the identification of potential Cox-2 inhibitors from selective coxibs." *Interdisciplinary Sciences: Computational Life Sciences* 11, no. 2 (2019): 153-169. [Impact Factor: 3.9]
- Daggupati, Trinath, **Rishika Pamanji**, and Suneetha Yeguvapalli. "In silico screening and identification of potential GSK3 β inhibitors." *Journal of Receptors and Signal Transduction* 38, no. 4 (2018): 279-289. [Impact Factor: 2.3]
- Daggupati, Trinath, Kumaraswamy Naidu Chitralla, **Rishika Pamanji**, and Suneetha Yeguvapalli. "Molecular screening and analysis of novel therapeutic inhibitors against c-Jun N-terminal kinase." *Medicinal Chemistry Research* 26, no. 9 (2017): 2112-2118. [Impact Factor: 3.1]
- E. Sreeramulu, K. Kranthi Kumar, **Rishika Pamanji**, B. Umadevi and Y. Suneetha, "Molecular docking studies of flavonoids against driver markers in low grade glioma: Exploring novel therapeutic approaches." *International Journal of Advanced Research*, 13, (2025): 849-852.
Google Scholar: <https://scholar.google.com/citations?user=JhDibq4AAAAJ&hl=en>

Papers presented at conferences: 15

EDUCATION

University of Houston 2026 – Present, USA, Ph.D. in Biomedical Engineering

San Jose State University 2023, USA, M.S. in Computer Science

Sri Venkateswara University 2019, India, B.E. in Computer Science

MS Thesis title: Advances in Robustness of Image-based Malware Detection, **Rishika Pamanji**, SJSU, 2023